J&J Supplier Rationalization: Business Case

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Executive summary

Recommendation: cut J&J's active supplier base by more than half and put every retained supplier on the agreement its category calls for. Tested on **real procurement data** (16,933 suppliers, $7.3B a year), the rules keep **4,648 suppliers (−73% of records, −59% of the active base)** while keeping **87% of spend** and **96.6% of active small/diverse spend** with retained suppliers.

Value at J&J scale (base case): $366M a year in run-rate savings for **$40M one-time cost**, a 3-year NPV of **$572M**. The original top-down estimate was $385M, so the headline holds, but the mix is different:

- **$86M a year is measured in the data**: a 5–10% price gap on spend moved from small suppliers in leverage and non-critical categories, plus fewer suppliers to maintain.
- **$280M depends on one assumption**: re-tendering the suppliers we keep in leverage categories. To reach the **$300M goal, that re-tender must save at least 1.15%**. That is the first thing to test in a pilot.

Strategic suppliers are never cut. Real J&J examples from SEC filings (Alkermes, MeiraGTx, Mersana) are sole-source manufacturers that move to partnership agreements; Emergent BioSolutions, whose contract J&J ended in 2022 over quality breaches, shows why critical products need a qualified second source.

Student case study: methods run on real public procurement data (State of California purchase orders, FY2012–15) and are scaled to J&J's published totals. J&J's internal supplier data is not public.

Goal and method

Goal: decide which of J&J's ~29,000 suppliers to keep, consolidate or exit, with a data reason behind every exit, and show the result saves at least $300M a year without adding supply risk.

We answer four questions, one method each. Each answer feeds the next; together they produce the exit list and the business case numbers.

#	Decision question	Method	Tool	Output
1	Where does the spend sit?	Spend cube + Pareto / ABC analysis	SQL	A / B / C segment per supplier
2	How should each category be handled?	Kraljic matrix (value vs supply risk)	Python	Quadrant per category, which sets the agreement type
3	Which suppliers stay and which go?	Duplicate detection + weighted scorecard + rules 1–5	Python (rapidfuzz), SQL	Keep / consolidate / exit list, one reason code per supplier
4	What is it worth?	Savings model, low / base / high scenarios	Python	Savings, one-time cost, NPV

The one AI/ML piece: fuzzy name matching to find duplicate suppliers (for example "Acme Lab Supply Inc" and "ACME Laboratory Supplies"). Duplicates are the easiest exits to justify, so this is where ML directly supports the recommendation.

One dashboard (Power BI or Tableau), one view per question: Pareto curve, Kraljic 2x2, exit list by reason code, savings bridge.

Deliberately left out, because none of them would change the recommendation:

Method	Why it is out
NLP spend classification	Our synthetic data is already categorized; it solves a real-data problem we don't have
Supplier risk prediction (random forest, XGBoost)	Needs real failure history; on synthetic data it would only learn our own assumptions
K-means clustering	The scorecard and Kraljic matrix already group suppliers in ways a reader can explain
Linear programming optimization	Rules 1–5 already encode the constraints; an optimizer adds complexity, not a different list
Monte Carlo simulation	Three scenarios show the range clearly enough

Data needed: five tables (suppliers, spend transactions, categories, scorecards, contracts), generated synthetically and calibrated to J&J's public totals.

Current state

J&J already runs a large, shrinking supplier base: about 29,000 suppliers in 2024, down from 38,500 in 2023, after the 2023 separation of its consumer business (Kenvue).

Metric	Value	Source
Suppliers, 2024	29,000+	2024 Health for Humanity Report
Suppliers, 2023	38,500+	2023 Health for Humanity Report
Spend under procurement process	~$35B across ~35,000 suppliers	J&J job posting, Procurement Global Process Owner
Tier 1 diverse supplier spend, 2022	$3B	J&J DEI in our supply base
Billion Dollar Roundtable member	13 consecutive years (2023)	2023 supplier diversity page

The often-quoted figure of 80,000 suppliers appears to be an older, pre-separation number. This case uses the current ~29,000 baseline; the model re-runs on any baseline.

The problem

Most of J&J's suppliers carry little spend but full overhead. In a typical Pareto-shaped base, about 85% of suppliers hold under 10% of spend, yet each one still needs onboarding, master data, compliance screening, invoices and payments.

- **Lost leverage:** the same category (lab supplies, MRO, IT services, travel, packaging) is bought from many vendors at many prices.
- **Process cost:** every active supplier costs an estimated ~$2,000 a year to maintain, before any audits.
- **Risk blind spots:** with tens of thousands of suppliers, risk monitoring (financial, ESG, cyber, sanctions) cannot go deep on each one.
- **Compliance exposure:** more suppliers means more third-party due diligence under anti-bribery and supply chain due-diligence rules.

The opportunity is not to squeeze strategic suppliers; it is to consolidate the long tail and the duplicated mid-tier.

Proposed approach

Segment suppliers by spend, then apply a different rule to each segment. Strategic suppliers are kept and deepened; the core is consolidated by category; the tail is moved to preferred vendors and catalogs.

Segment	Suppliers	Spend ($M)	Share of spend	Target cut	Suppliers after	Treatment
A Strategic	580	24,500	70%	0%	580	Partnership agreements, joint value engineering
B Core	3,770	7,700	22%	25%	2,828	Consolidate per category to 2–3 preferred suppliers
C Tail	24,650	2,800	8%	60%	9,860	Move to preferred vendors, catalogs, marketplace
Total	**29,000**	**35,000**	**100%**	**54%**	**13,268**	

Rationalization rules (applied supplier by supplier in the data phase):

1. **Never exit a regulated sole source** (FDA-validated API, excipient, component or contract manufacturer) without a qualified alternative and a requalification plan.
2. **Keep at least two qualified sources** for anything patient-critical.
3. **Protect diverse and small suppliers:** they are consolidated into preferred-supplier and Tier 2 programs, not dropped, so Tier 1 diverse spend stays at or above $3B.
4. **Exit first** where a supplier is inactive (no PO in 18 months), duplicated (same category, same site, lower score) or below a minimum spend threshold.
5. **Score the rest** on cost, quality, delivery, risk, ESG and diversity; the lowest-scoring supplier in a category exits first.

What the real data shows

The approach was tested on 344,504 real State of California purchase lines (FY2012–15). After removing pass-through and procurement-exempt spend, 278,632 lines remain: **$21.9B over 3 years across 16,933 suppliers**. Every assumption above held or was corrected.

Step	Question	Result on real data	vs. assumption
2 Pareto / ABC (SQL)	Where does the spend sit?	Top 1.9% of suppliers = 70% of spend; top 10.7% = 92%	Assumed 2% / 15%: **holds**
2 Spend cube	How much of the tail is dead weight?	59% of tail suppliers are inactive 18+ months or one-time vendors	Assumed 60% tail cut: **holds**
3 Kraljic	How should each category be handled?	Leverage 59% of spend (20 categories); strategic 21%; bottleneck 13%; non-critical 7% but 21,991 supplier relationships	Sets the agreement type per category
4 Duplicates (ML) + rules	Which suppliers stay?	858 duplicate records found (TF-IDF name matching); 4,648 kept, 8,824 exited, 2,623 consolidated, 838 merged	Assumed −54%: real −59% of active base
5 Price-gap test	Do exiting suppliers charge more?	Same goods item: +9.8% in leverage, +5.2% in non-critical, but **−10% in bottleneck** and −7% in strategic	Assumed 5–8% everywhere: **corrected**

The key correction: consolidation only saves money in leverage and non-critical categories. In bottleneck and strategic categories the small suppliers were cheaper, so the plan secures supply there instead of squeezing price.

Real J&J strategic suppliers

Read from the suppliers' own SEC 10-K filings:

Supplier	Supplies J&J with	Lesson
Alkermes	Sole manufacturer of RISPERDAL CONSTA (Ohio plant)	Sole-source, regulated: never cut; partnership MOU + supply and quality agreements

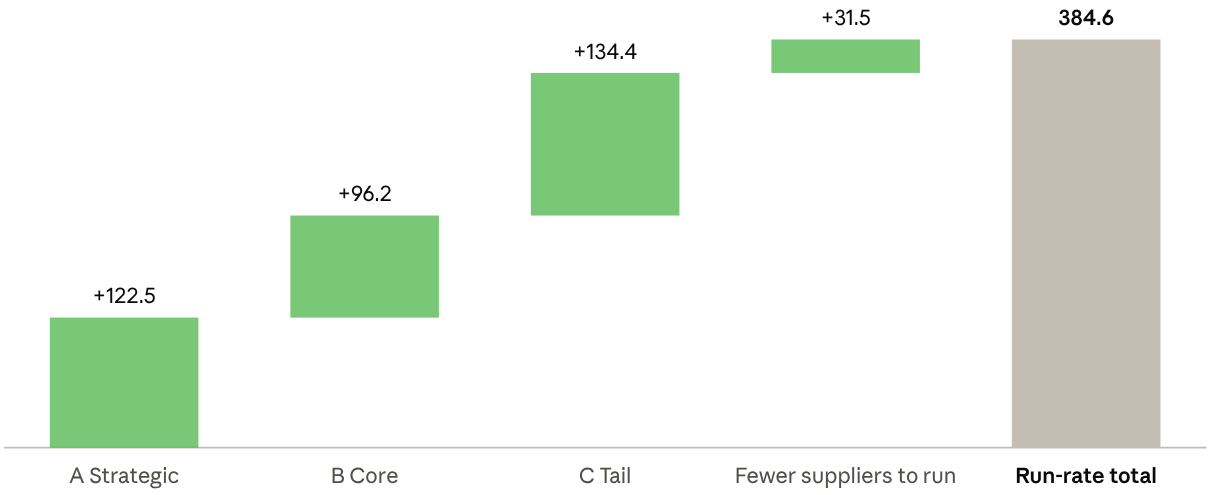
Supplier	Supplies J&J with	Lesson
MeiraGTx	RPGR gene therapy, 4+1-year supply agreement (Dec 2023)	Long-term supply agreement, not a tender
Mersana	Clinical ADC manufacturing, with a technology-transfer right for J&J	A second-source path written into the contract
Emergent BioSolutions (former)	COVID-19 vaccine drug substance; J&J ended it in 2022 over GMP breaches, settled for $50M in 2024	Why a qualified second source is a rule, not an option

Financial case

Original top-down estimate (before the real data): $385M a year at full run-rate against $59M one-time cost. The real-data model below replaces it.

Tail consolidation is the largest single source of the run-rate savings

Annual run-rate savings, $M, base case (year 3 onward). Green = savings added; grey = total.



Business case model (model.py), base case assumptions

Price savings make up $353M of the total; the remaining $31.5M comes from no longer maintaining ~15,700 exited suppliers.

One-time cost: $58.6M, made up of program team, analytics and tools ($15M), contracting and legal ($5M), supplier transition and requalification ($31M) and a 15% contingency. Core-segment exits carry most of the transition cost ($25k each) because some need quality requalification.

Scenarios (savings ramp 30% / 70% / 100% over years 1–3, 8% discount rate):

Scenario	Savings vs base	Cost vs base	Run-rate ($M/yr)	3-yr savings ($M)	3-yr NPV ($M)	ROI
Low	50%	125%	192	385	248	4.3x
Base	100%	100%	385	769	584	12.1x
High	130%	90%	500	1,000	783	18.0x

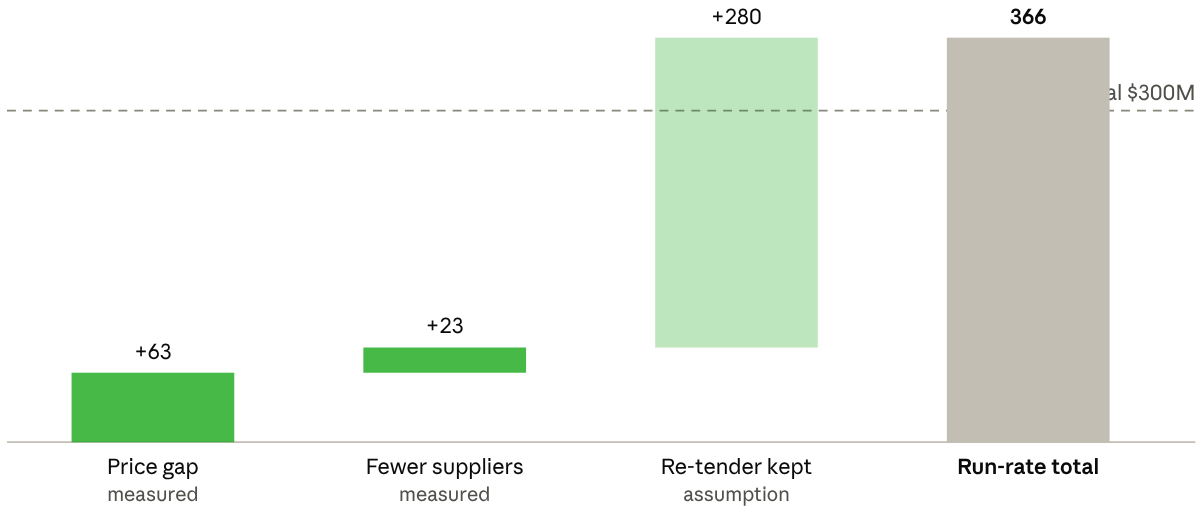
Run-rate savings of 1.1% of addressable spend sit at the conservative end for a consolidation program, which is what makes the case defensible.

Real-data savings model (replaces the estimate)

Each step-4 decision is tied to one lever, using the price gap measured in the data where it exists. Inactive suppliers count as $0 because their spend had already stopped.

Only $86M of the $366M base case is measured in the data

J&J run-rate savings, $M per year, base case. Solid = measured in real data; pale = assumption.



scripts/05_savings.py on the real supplier list, scaled to J&J ($35B spend, 29,000 suppliers)

J&J scale, $M	Low	Base	High
Price gap on re-routed spend (measured)	0	63	126
Fewer active suppliers to maintain	11	23	34
Re-tender of kept leverage suppliers (assumption)	93	280	561
Run-rate savings per year	**105**	**366**	**721**
One-time cost	40	40	40
3-year NPV (8%, ramp 30/70/100%)	135	572	1,165
Meets $300M goal?	No	Yes	Yes

What has to be true: the re-tender of kept leverage suppliers must save **at least 1.15%** for the base case to clear $300M. The MSA template's volume tiers and benchmark clause are how that saving is captured and measured.

Agreements for retained suppliers

Each retained supplier gets the agreement its role needs. An MOU fits strategic partners because it records shared intent, but it is not binding on price or supply, so it always sits on top of a binding contract.

Kraljic quadrant	Typical J&J categories	Agreement	What it locks in	Term
Strategic (high value, high risk)	APIs, biologics inputs, contract manufacturers	MOU / strategic partnership + supply agreement + quality agreement	Joint roadmap, capacity reservation, quality and change-control terms	3–5 yrs
Bottleneck (low value, high risk)	Specialty excipients, single-source components	Long-term supply agreement + dual-sourcing plan	Safety stock, notice of discontinuation, second source qualification	3–5 yrs

Kraljic quadrant	Typical J&J categories	Agreement	What it locks in	Term
Leverage (high value, low risk)	Packaging, lab supplies, IT services, logistics	Master supply / services agreement (MSA)	Volume-tiered pricing, SLAs, rebates, payment terms	2–3 yrs
Non-critical (low value, low risk)	MRO, office supplies, travel, small services	Catalog / marketplace / P-card	Fixed catalog pricing, punch-out ordering	1–2 yrs

Regulated suppliers also keep their GMP quality agreements regardless of commercial terms. In the project phase, I will draft a template MOU and an MSA term sheet.

Templates built (step 7), one per quadrant, applied to the step-4 keep list:

Template	Applies to	Suppliers
Strategic partnership MOU	Strategic	417
MSA term sheet with volume tiers and benchmark clause	Leverage	1,648
Long-term supply agreement with second-source plan	Bottleneck	1,061
Catalog / purchasing-card terms	Non-critical	1,522
Transition notices (closure, consolidation, duplicate merge)	Not kept	12,285

The clauses mirror real J&J contracts read in the SEC filings: notice periods (Alkermes), fixed term with extension option (MeiraGTx) and technology transfer (Mersana). Full text: docs/agreements.

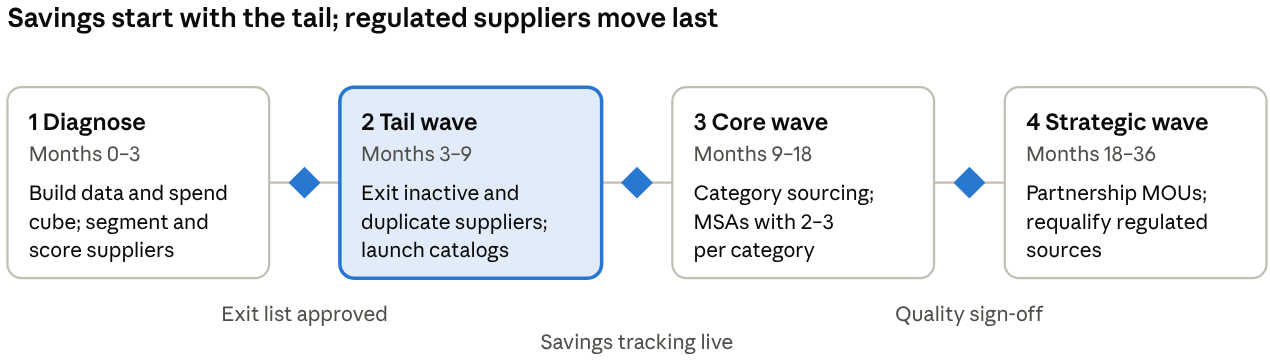
Risks and guardrails

The largest risk is supply continuity in regulated categories; the rules above exist mainly to contain it.

Risk	Impact	Guardrail
Supply disruption after exiting a source	Patient supply, revenue	Rules 1–2: no regulated sole-source exits; dual sourcing for critical items
Requalification takes longer than planned	Savings delayed 12–24 months	Regulated changes scheduled in later waves; savings ramp assumes delay
Breaking diversity commitments	Reputation, Billion Dollar Roundtable status	Rule 3: diverse spend floor of $3B tracked monthly
Over-concentration on fewer suppliers	Pricing power shifts to suppliers	Max share of wallet per category; supplier financial health monitoring
Savings leakage (maverick buying)	Benefits not realized	Catalog and PO controls; spend compliance KPI
Supplier relationship damage	Credo values, goodwill	Clear exit notices, transition support, 90-day wind-down

Roadmap and KPIs

The rollout runs 36 months in four waves, ordered from lowest risk to highest so savings fund the harder work.



Rollout plan · 4 phases, 3 gates (diamonds)

KPI	Baseline	Year 3 target
Active suppliers	~29,000	~11,900 (−59% of active base)
Run-rate savings	$0	$366M a year (base case)
Spend under contract (MOU / MSA / catalog)	Not public	95% of addressable spend
Tier 1 diverse supplier spend	$3B (2022)	⩾ $3B
Critical items with 2+ qualified sources	Not public	100%
Supply disruptions caused by exits	n/a	0

Deliverables

Deliverable	Where
Code, SQL, step write-ups, agreements	GitHub repo
All step outputs with live formulas and a Files_and_Sources tab	Outputs workbook
4-page Power BI dashboard (21 DAX measures)	powerbi/
Verified J&J suppliers from SEC filings	docs/07_real_jnj_suppliers.md
Raw data to decision trace (344,504 rows)	Rebuilt with scripts/build_data_workbook.py (100 MB, outside git)

Assumptions and sources

All figures not linked below are modeling assumptions, chosen to be conservative and easy to change.

Assumption	Value
Addressable spend	$35B
Suppliers	29,000

Assumption	Value
Segment split (suppliers / spend)	A 2% / 70%, B 13% / 22%, C 85% / 8%
Price savings on addressed spend	A 0.5%, B 5%, C 8%
Annual cost to maintain one supplier	$2,000
Transition cost per exited supplier	B $25,000, C $500
Program + legal	$15M + $5M, plus 15% contingency
Savings ramp	30% / 70% / 100% (years 1–3)
Discount rate	8%

Sources:

- J&J 2024 Health for Humanity Report
- J&J 2023 Health for Humanity Report
- J&J careers: Director, Procurement Global Process Owner
- J&J: DEI in our supply base (2022)
- J&J 2023: Supplier diversity and inclusion